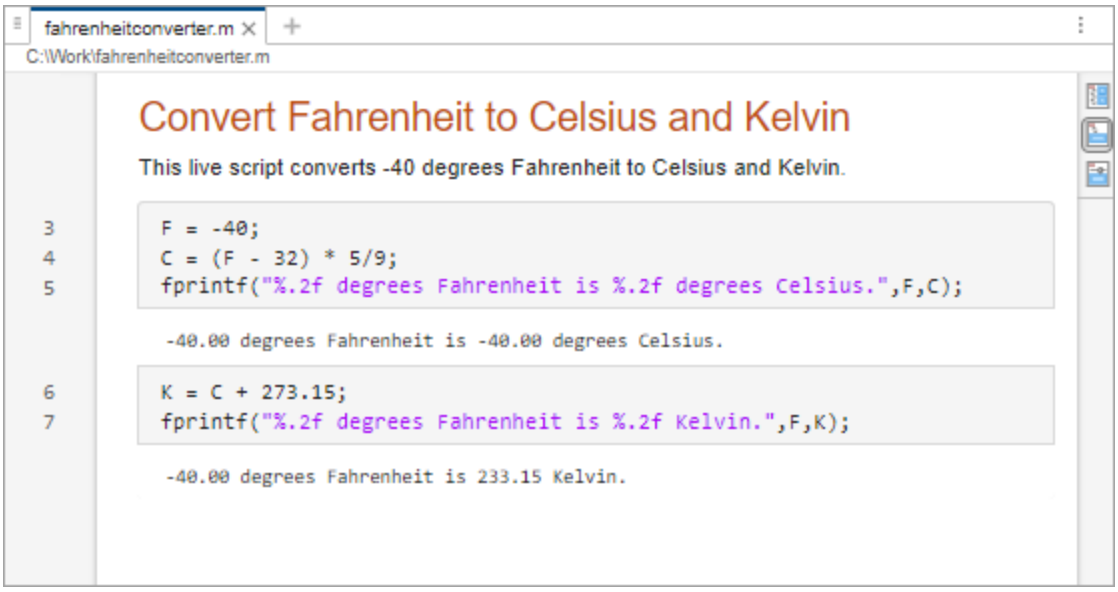
Live Code File Format (`.m`)

Since R2025a

Starting in R2025a, the Live Editor supports a new plain text Live Code file format (`.m`) for live scripts as an alternative to the default binary Live Code file format (`.mlx`). The plain text format is based on the standard MATLAB® code file format (`.m`) and supports all Live Editor features, including output, formatted text, interactive controls, and tasks. Live scripts saved in this format open in the Live Editor and behave like other live scripts.



Benefits to Plain Text Live Code File Format (`.m`)

There are several benefits to saving live scripts using the new plain text Live Code file format:

- Files open in external code editors — You can open live scripts saved using the plain text Live Code file format in any external text or code editor that supports plain text files.
- Increased transparency — Files saved using the plain text file Live Code format do not trigger security concerns. One exception might be if the live script contains images and plots, which are saved as Base64 (ASCII-binary) strings in the plain text file.
- Improved integration with source control — You can use external source control tools to compare and merge live scripts saved using the plain text file Live Code format.

Save Live Scripts as Plain Text

To save a live script using the plain text Live Code file format (`.m`):

1. Go to the **Live Editor** tab, and in the **File** section, select **Save** > **Save As**.
2. Enter a name for your live script. If you are saving an existing live script using the plain text Live Code file format, enter a different name than the original filename to avoid shadowing issues.
3. Select `MATLAB Live Code File (UTF-8) (*.m)` as the **Save as type**.
4. Click **Save**.

By default, live scripts saved using the plain text Live Code file format (`.m`) open in the Live Editor. To open the file as plain text in the Editor, right-click the file in the Files panel and select **Open as Text**.

> **Note**
>
> Editing a live script as plain text might prevent it from opening correctly in the Live Editor.

Change the Default Live Script File Format

By default, new live scripts are saved using the binary Live Code file format (`.mlx`). To make the plain text Live Code file format the default file format for live scripts, go to the **Home** tab, and in the **Environment** section, click **Settings**. Select **MATLAB** > **Editor/Debugger**, and in the **Live script format** section, select `M` as the **Default live script file format**.

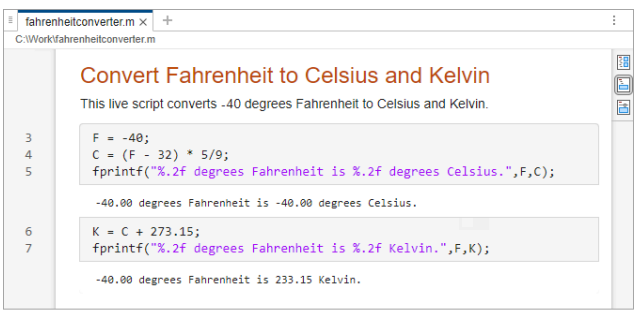
Structure of Plain Text Live Code File

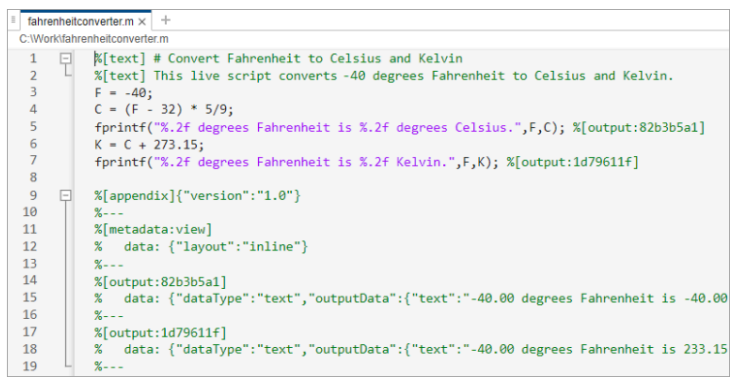
The Live Editor uses custom markup to save formatted text, generated output, interactive controls, and tasks as plain text. The Live Editor saves the custom markup in two places within the live script file:

- Alongside the code as inline markup
- At the bottom of the file in an appendix

For example, suppose that you have a live script, `fahrenheitconverter.m`, saved using the plain text Live Code file format. By default, when you open `fahrenheitconverter.m`, the file opens in the Live Editor. To open `fahrenheitconverter.m` as plain text in the Editor, right-click the file in the Files panel and select **Open as Text**.

Notice that in both the Live Editor and Editor, the line number for the first code line is 3. The reason is that in live scripts saved using the plain text Live Code file format, each line of text accounts for one or more code lines in the saved file. In addition, because lines of text are saved as code comments in the file, inserting text between statements that span multiple code lines is not supported.





Inline Markup

The Live Editor uses inline markup to save text items, such as formatted text, tables, equations, and hyperlinks. For example, this live script contains markup for a title line and a normal text line.

```
1  %[text] # Convert Fahrenheit to Celsius and Kelvin
2  %[text] This live script converts -40 degrees Fahrenheit to Celsius and Kelvin.
3  F = -40;
4  C = (F - 32) * 5/9;
5  fprintf("%.2f degrees Fahrenheit is %.2f degrees Celsius.",F,C); %[output:82b3b5a1]
6  K = C + 273.15;
7  fprintf("%.2f degrees Fahrenheit is %.2f Kelvin.",F,K); %[output:1d79611f]
8
9  %[appendix]{"version":"1.0"}
10 %---
11 %[metadata:view]
12 %   data: {"layout":"inline"}
13 %---
14 %[output:82b3b5a1]
15 %   data: {"dataType":"text","outputData":{"text":"-40.00 degrees Fahrenheit is -40.00 degrees Celsius.","truncated":false}}
16 %---
17 %[output:1d79611f]
18 %   data: {"dataType":"text","outputData":{"text":"-40.00 degrees Fahrenheit is 233.15 Kelvin.","truncated":false}}
19 %---
```

Appendix

The Live Editor uses an appendix at the end of the file to save larger amounts of data that cannot be saved inline and to save file information that is not linked to a specific line number. For example, this live script contains an appendix with markup that saves the current view of the live script, as well as the two outputs that the live script creates.

```
%[text] # Convert Fahrenheit to Celsius and Kelvin
%[text] This live script converts -40 degrees Fahrenheit to Celsius and Kelvin.
F = -40;
C = (F - 32) * 5/9;
fprintf("%.2f degrees Fahrenheit is %.2f degrees Celsius.",F,C); %[output:82b3b5a1]
K = C + 273.15;
fprintf("%.2f degrees Fahrenheit is %.2f Kelvin.",F,K); %[output:1d79611f]

%[appendix]{"version":"1.0"}
%---
%[metadata:view]
%   data: {"layout":"inline"}
%---
%[output:82b3b5a1]
%   data: {"dataType":"text","outputData":{"text":"-40.00 degrees Fahrenheit is -40.00 degrees Celsius.","truncated":false}}
%---
%[output:1d79611f]
%   data: {"dataType":"text","outputData":{"text":"-40.00 degrees Fahrenheit is 233.15 Kelvin.","truncated":false}}
%---
```

Reference IDs

For features that require additional markup, the Live Editor uses inline markup with a reference ID to save basic feature information inline and link to additional markup in the appendix. When merging live scripts that contain differences in reference IDs, make sure that the inline reference ID and the reference ID in the appendix match in the merged result.

```
%[text] # Convert Fahrenheit to Celsius and Kelvin
%[text] This live script converts -40 degrees Fahrenheit to Celsius and Kelvin.
F = -40;
C = (F - 32) * 5/9;
fprintf("%.2f degrees Fahrenheit is %.2f degrees Celsius.",F,C); %[output:82b3b5a1]
K = C + 273.15;
fprintf("%.2f degrees Fahrenheit is %.2f Kelvin.",F,K); %[output:1d79611f]

%[appendix]{"version":"1.0"}
%---
%[metadata:view]
%   data: {"layout":"inline"}
%---
%[output:82b3b5a1]
%   data: {"dataType":"text","outputData":{"text":"-40.00 degrees Fahrenheit is -40.00 degrees Celsius.","truncated":false}}
%---
%[output:1d79611f]
%   data: {"dataType":"text","outputData":{"text":"-40.00 degrees Fahrenheit is 233.15 Kelvin.","truncated":false}}
%---
```

Markup Details

In live scripts saved using the plain text Live Code file format, the Live Editor saves custom markup as comments. In general, the Live Editor uses Markdown to save text items, such as formatted text, tables, images, and hyperlinks, and LaTeX commands to save equations.

This table provides more information about the custom markup used in the plain text Live Code file format (`.m`).

Markup Type	Markup Syntax

Formatted text	Text is saved inline using the `%[text]` markup. Markup within the line specifies the formatting.
	`%[text] This is basic text`
	`%[text] # This is a title`
	`%[text] ## This is a header`
	`%[text] This is **bold**, *italic*, `monospace`, and <u>underlined</u>`
	`%[text]{"align":"center"} This block of text is center aligned`

Markup Type	Markup Syntax
Table of contents	Table of contents is saved inline using the `%[text:tableOfContents]` markup. `%[text:tableOfContents]{"heading": "Table of Contents"}`
Code example	Code examples are saved inline using the `%[text]` markup. Plain code example: `%[text] ```` `%[text] x = 1;` `%[text] y = "hello";` `%[text] ```` MATLAB code example: `%[text] ```matlabCodeExample` `%[text] x = 1;` `%[text] y = "hello";` `%[text] ````

Table	Tables are saved inline using the `%[text:table]` markup.
	`%[text:table]`
	`%[text] | abc | def |`
	`%[text] | --- | --- |`
	`%[text] | 123 | 456 |`
	`%[text:table]`

Image	Images are saved both inline and in the appendix.
	
	Syntax inline:
	`%[text] this image: ![`*alt text*`](text:image:`*imageid*`)`
	
	Syntax in appendix:
	`%[text:image:`*imageid*`]`
	`%  data: { ... }`
	`%---`

Hyperlink	Hyperlinks are saved inline using the `%[text]` markup.
	`%[text] web page: [MathWorks Website](https://mathworks.com)`
	`%[text] file: [Duck Duck Go](/path/to/file.txt)`

Equation	Equations are saved as LaTeX commands inline using the `%[text]` markup. `%[text] $x = 4^2${"altText": "equation alt text"}`

Output	Output is saved using the `%[output]` markup both inline and at the bottom of the file in the appendix.
	
	Syntax inline:
	`%[output:`*outputid*`]`
	
	Syntax in appendix:
	`%[output:`*outputid*`]`
	`%  data: { ... }`
	`%---`

Markup Type	Markup Syntax

Control	Controls are saved using the `%[control]` markup both inline and in the appendix. The `{"position":[start,end]}` markup defines the start position and end position of the control within the code line.
	
	Syntax inline:
	`%[control:`*`controltype`*`:`*`controlid`*`]{"position":[14,16]}`
	
	Syntax in appendix:
	`%[control:`*`controltype`*`:`*`controlid`*`]`
	`%  data: { ... }`
	`%---`

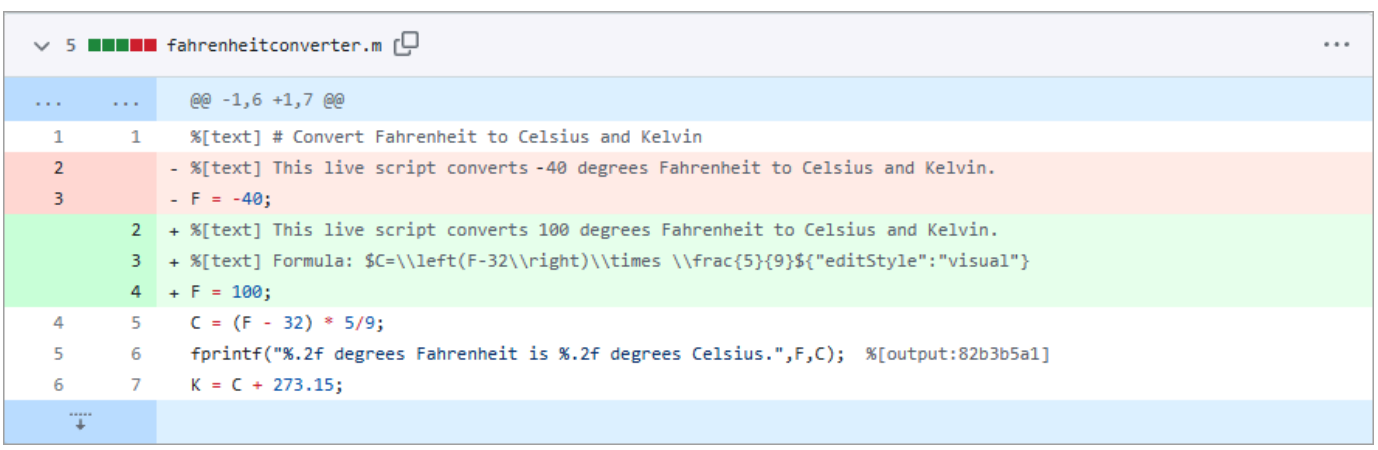
Live Editor task	Live Editor tasks are saved using the `%[task]` markup both inline next to each task code line and in the appendix.
	
	Syntax inline:
	`%[task:`*`taskid`*`]`
	
	Syntax in appendix:
	`%[task:`*`taskid`*`]`
	`%  data: { ... }`
	`%---`

Live script view	The current view of the live script is saved using the `%[metadata]` markup in the appendix.
	`%[metadata:view]`
	`%  data: {"layout":"inline","rightPanelPercent":40}`
	`%---`

Source Control

When using external source control tools, you can compare and merge live scripts saved using the plain text Live Code file format (`.m`) directly from the tool. Live scripts saved using the binary Live Code file format (`.mlx`) can only be compared using the MATLAB Comparison Tool.

For example, in GitHub®, when you add a live script saved using the plain text Live Code file format to a repository, you can see the differences between each revision of the file.



Source Control Limitations and Workarounds

Some external source control tools have limitations around file size that might prevent the comparison of live scripts saved using the plain text Live Code file format:

- GitLab® and GitHub throw an error and do not display the differences in a file if the file size is too big.
- Perforce® tries to display the differences but might become unresponsive if the file size is too big. To avoid Perforce becoming unresponsive, you can try disabling the Show Inline Differences option.

For the best results, try using GitHub Desktop or Visual Studio Code. You also can reduce file size by disabling the saving of output to your files. To disable saving output, go to the **Home** tab, and in the **Environment** section, click **Settings**. Select **MATLAB** > **Editor/Debugger** > **Saving** and in the **Output Save Options** section, clear **Save output to file**. Alternatively, clear all output before submitting your files to source control.

For more information, see the documentation for your source control tool.

See Also

Topics

What Is a Live Script or Function?

Create Live Scripts in the Live Editor

Live Code File Format (.mlx)